Big Data Analysis Report

Analysis of Diabetes Dataset Using Apache Spark

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1. Introduction

This report presents a comprehensive Big Data workflow applied to the Diabetes CSV dataset using Apache Spark, one of the most powerful distributed processing engines. The dataset contains essential medical diagnostic attributes such as glucose level, BMI, age, insulin, and blood pressure, which are commonly used for early diabetes risk identification.

The main objective of this work is to demonstrate how Spark handles large-scale data ingestion, cleaning, transformation, and analytics efficiently. By distributing computations across multiple processing units, Spark enables faster execution, automatic parallelization, and scalable processing, which makes it suitable for real-world medical and healthcare applications.

2. Methodology

The workflow followed four main phases :

• Data Ingestion

The diabetes.csv file was loaded into a Spark DataFrame, allowing schema inspection and efficient handling of numerical medical attributes. Spark's lazy evaluation and optimized execution engine make the ingestion step capable of handling much larger datasets than traditional tools.

• Data Cleaning

Several medical fields contain zero values that represent missing or invalid measurements. Therefore, the following columns were selected for cleaning: Glucose, BloodPressure, SkinThickness, Insulin, and BMI.

Zero entries were replaced with NULL, and rows containing missing values were removed. This step ensured that the final analysis reflects realistic medical patterns.

• Data Transformation

Two engineered columns were added to support deeper analysis:

• AgeGroup: Categorizes patients into meaningful clinical groups such as Under 30, 30–39, 40–49, and 50+.

• GlucoseLevel: Categorizes glucose values into Normal, Prediabetic, and DiabeticRange.

These transformations enabled demographic and clinical comparisons that are difficult to observe in raw data.

• Data Analysis

Spark's groupBy and aggregation functions were used to compute:

• Distribution of diabetes outcomes

• Average glucose, BMI, and age per group

• Demographic trends across age categories

• Glucose-level distributions across diabetic and non-diabetic patients

The analysis highlights patterns that are medically meaningful and consistent with known diabetes risk factors.

3. Results

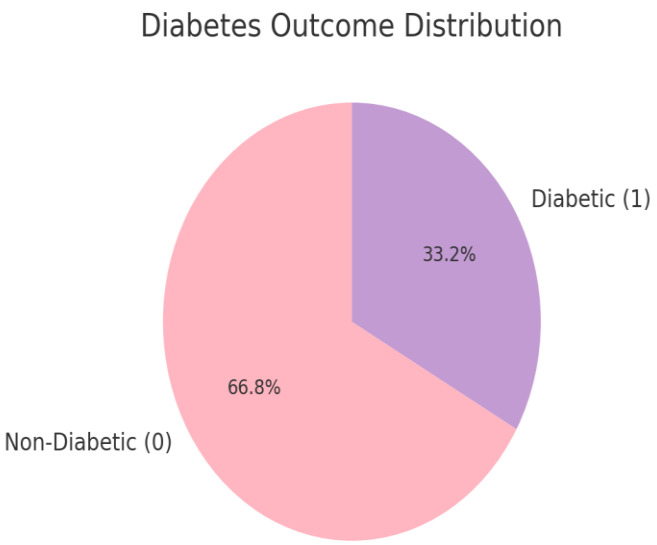
The visual analysis highlights the key patterns in the diabetes dataset.

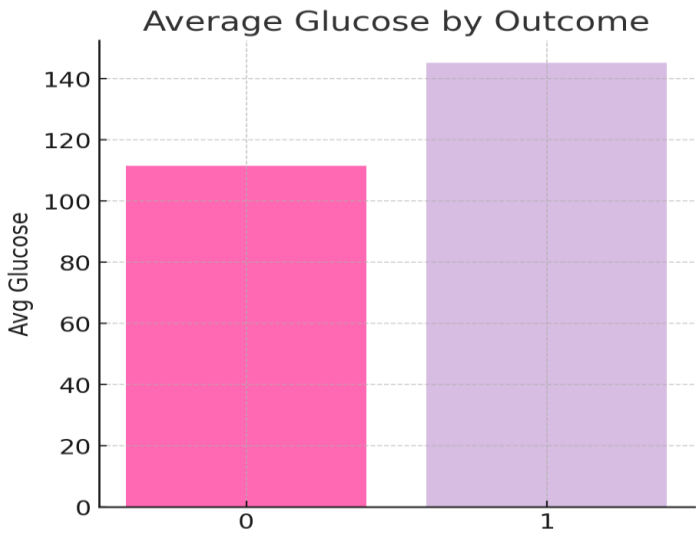
The Diabetes Outcome Distribution chart shows that most patients are non-diabetic (66.8%), while diabetic cases represent 33.2%.

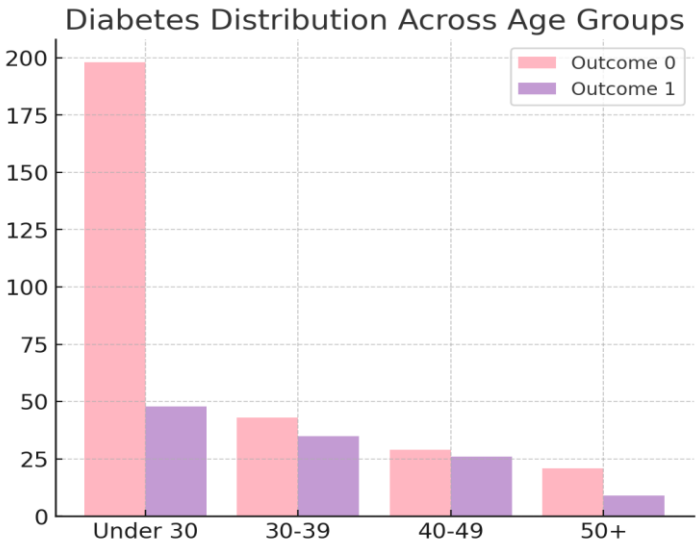
The Average Glucose by Outcome chart demonstrates a clear difference between groups: diabetic patients have substantially higher glucose levels than non-diabetic individuals, confirming the clinical relationship between high glucose and diabetes.

The Age Group Distribution chart shows that diabetes becomes more common in older age groups, particularly among individuals aged 40–49 and 50+, while younger individuals (Under 30) show the lowest risk.

Note: Additional tables—such as missing-value statistics, summary statistics, and detailed group comparisons—are provided in the code section.







4. Conclusion

The Big Data analysis performed using Apache Spark successfully uncovered meaningful medical and demographic patterns within the diabetes dataset. By combining data cleaning, transformation, and distributed analytics, Spark demonstrated its effectiveness in producing scalable and accurate clinical insights.

The engineered AgeGroup and GlucoseLevel features provided clearer visualization of risk trends, while the statistical comparisons confirmed the relationship between glucose level, BMI, age, and diabetes outcomes.

Overall, Spark proved to be an efficient and powerful tool for processing large medical datasets and supporting data-driven healthcare analytics.